

ECN 594: Consumer Surplus, Monopoly Pricing, and Price Discrimination

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Plan for today

1. **Consumer surplus**
2. Welfare from new products: applications
3. Monopoly pricing: using estimated elasticities
4. Monopoly regulation

Why do we care about consumer surplus?

- Policy analysis requires measuring welfare
- Questions we want to answer:
 - How much do consumers gain from a new product?
 - How much do consumers lose from a merger?
 - What is the welfare cost of a price increase?
- Need a way to compute consumer surplus from our demand model

Consumer surplus in logit: the log-sum formula

- For consumer i in market t , expected utility from choosing among J products:

$$E[\max_j u_{ijt}] = \ln \left[\sum_{j=0}^J \exp(\delta_{jt} + \mu_{ijt}) \right] + \text{constant}$$

- This is the “log-sum” or “inclusive value”
- Consumer surplus (in dollars):

$$CS_{it} = \frac{1}{|\alpha_i|} \ln \left[\sum_{j=0}^J \exp(\delta_{jt} + \mu_{ijt}) \right]$$

- Divide by $|\alpha_i|$ (consumer i 's price coefficient) to convert to dollars

Intuition for the log-sum

- Think of choosing the best option as a lottery
- Each product gives you a random utility draw
- **Expected value of the BEST draw** is the log-sum
- Key insights:
 - More options $\rightarrow$ higher CS (more lottery tickets)
 - Better options $\rightarrow$ higher CS (higher δ_j)
 - Higher price sensitivity $\rightarrow$ divide by larger $|\alpha|$

Worked example: CS change from removing a product

- **Question:** Market has 2 products with $\delta_1 = 1$, $\delta_2 = 0.5$. Outside option has $\delta_0 = 0$. Suppose $\alpha = -0.5$.
- What is the consumer surplus loss if product 1 is removed?

Take 3 minutes to solve this.

Worked example: CS change (solution)

- **Solution:**

- **Before removal:**

$$CS^{\text{before}} = \frac{1}{|\alpha|} \ln(e^0 + e^1 + e^{0.5}) = 2 \ln(1 + 2.72 + 1.65) = 2 \ln(5.37) = 3.36$$

- **After removal:**

$$CS^{\text{after}} = \frac{1}{|\alpha|} \ln(e^0 + e^{0.5}) = 2 \ln(1 + 1.65) = 2 \ln(2.65) = 1.95$$

- **Loss:** $3.36 - 1.95 = 1.41$ dollars per consumer

Welfare from new products: Red Bus / Blue Bus

- The log-sum formula lets us value welfare from new products
- But there's a problem: the **Red Bus / Blue Bus** example
- **Setup:** Consumers choose how to commute
 - Choices: Car, Red Bus
 - Half choose Car, half choose Red Bus
 - So: $\delta_{\text{car}} = \delta_{\text{red bus}} = 0$

Red Bus / Blue Bus: introducing a new option

- Artificially introduce a **Blue Bus**
- But consumers are **color-blind!**
- Blue Bus is identical to Red Bus in every way
- Now: half choose Car, rest split between buses
- **Reality:** Welfare should NOT change
 - It's the same bus, just different color
 - No real new option for consumers

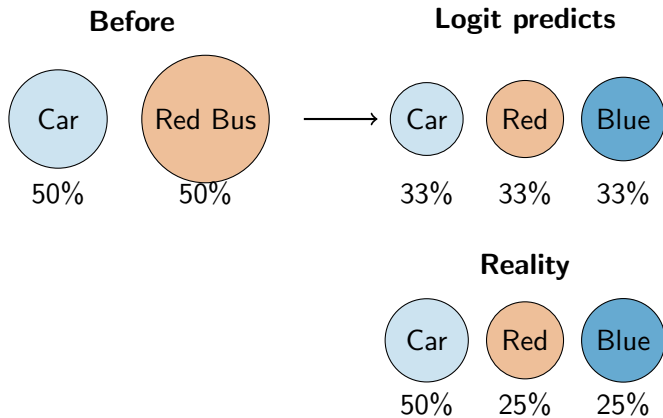
What does logit predict?

- Suppose we only observe data **pre-introduction** of Blue Bus
- **Pre-introduction:**
 - Normalize: $\delta_{\text{car}} = 0$. Then $\delta_{\text{red bus}} = 0$ (since equal shares)
 - Inclusive value: $\ln(e^0 + e^0) = \ln(2)$
- **Post-introduction** (using logit to predict):
 - $\delta_{\text{blue bus}} = \delta_{\text{red bus}} = 0$ (same bus)
 - Logit predicts: $s_{\text{car}} = s_{\text{blue}} = s_{\text{red}} = 1/3$
 - Inclusive value: $\ln(e^0 + e^0 + e^0) = \ln(3)$
- Logit says welfare **increased!** But nothing real changed...

Red Bus / Blue Bus: what went wrong?

- **Main issue:** We get an extra logit draw when introducing Blue Bus
- Logit gives an extra “lottery ticket” for each new product
- It doesn't know that buses are close substitutes

Red Bus / Blue Bus: Visual



Solution: use observed post-introduction shares

- **Key insight:** If we observe the market *after* introducing Blue Bus, we can correct this
- **Observed shares:** $s_{\text{car}} = 0.5$, $s_{\text{blue}} = s_{\text{red}} = 0.25$
- Back out δ from observed shares:

$$\delta_{\text{blue bus}} = \delta_{\text{red bus}} = \ln(0.25) - \ln(0.5) = \ln(0.5)$$

- **Corrected inclusive value:**

$$\ln(e^0 + 2 \times e^{\ln(0.5)}) = \ln(1 + 2 \times 0.5) = \ln(2)$$

- This is the **correct answer** — no welfare change!

Key takeaways: welfare from new products

- **Problem:** Introducing new products adds extra logit draws, biasing welfare up
- **Solution:** Observing post-introduction market shares can correct for this
 - Fitting the shares exactly means the extra logit gain is offset by lower δ 's
- **Bottom line:** Be careful computing welfare from hypothetical new products
 - If you observe the new product in data, fit the shares first
 - If purely counterfactual, results may be biased

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Application: the minivan (Petrin, 2002)

- **Question:** What are the welfare benefits of the minivan?
- Minivan:
 - Box-like vehicle first introduced in 1984
 - Increased height, decreased floor → easier entry
 - Extremely popular with families
- Motivation: New products are a key source of economic growth. How should we value them?



Petrin (2002): Main takeaways

- Beyond valuing the minivan, key methodological insights:
 1. Logit model performs poorly compared to mixed logit
 2. Use of **micro-data** to improve estimates
 3. Dealing with the Red Bus / Blue Bus problem

Petrin (2002): Model and data

- Utility: $u_{ijt} = x_{jt}\beta_{it} + \alpha_{it} \ln(y_i - p_{jt}) + \xi_{jt} + \epsilon_{ijt}$
- Price coefficient varies by income
- “Minivan” enters as a product characteristic
- Data: US market 1981–1993 (916 vehicles)
 - Prices, quantities, product characteristics
 - **Key**: CEX auto supplement links demographics $\leftrightarrow$ purchases

Petrin (2002): What is micro-data?

- **Micro-data** = individual-level survey data linking demographics to purchases
- Example: Consumer Expenditure Survey (CEX) auto supplement
 - Observe: household income, family size, vehicle purchased
 - Can compute: “What fraction of minivan buyers have 3+ kids?”

Petrin (2002): Demand estimates

TABLE 4
PARAMETER ESTIMATES FOR THE DEMAND-SIDE EQUATION

Variable	OLS Logit (1)	Instrumental Variable Logit (2)	Random Coefficients (3)	Random Coefficients and Microdata (4)
A. Price Coefficients (α 's)				
α_1	.07 (.01)**	.13 (.01)**	4.92 (9.78)	7.52 (1.24)**
α_2			11.89 (21.41)	31.13 (4.07)**
α_3			37.92 (18.64)**	34.49 (2.56)**

Petrin (2002): Substitution patterns

	B. Base Coefficients (β 's)			
Constant	-10.03 (.32)**	-10.04 (.34)**	-12.74 (5.65)**	-15.67 (4.39)**
Horsepower/weight	1.48 (.34)**	3.78 (.44)**	3.40 (39.79)	-2.83 (8.16)
Size	3.17 (.26)**	3.25 (.27)**	4.60 (24.64)	4.80 (3.57)*
Air conditioning standard	-.20 (.06)**	.21 (.08)**	-1.97 (2.23)	3.88 (2.21)*
Miles/dollar	.18 (.06)**	.05 (.07)	-.54 (3.40)	-15.79 (.87)**
Front wheel drive	.32 (.05)**	.15 (.06)**	-5.24 (3.09)	-12.32 (2.36)**
Minivan	.09 (.14)	-.10 (.15)	-4.34 (13.16)	-5.65 (.68)**
Station wagon	-1.12 (.06)**	-1.12 (.07)**	-20.52 (36.17)	-1.31 (.36)**
Sport-utility	-.41 (.09)**	-.61 (.10)**	-3.10 (10.76)	-4.38 (.41)**
Full-size van	-1.73 (.16)**	-1.89 (.17)**	-28.54 (235.51)	-5.26 (1.30)**
% change GNP	.03 (.01)**	.03 (.01)**	.08 (.02)**	.24 (.02)**

Petrin (2002): Counterfactual

- **Exercise:** Eliminate minivan, compute welfare change
- **Issue:** utility is not quasilinear
- Uses **Compensating Variation (CV)**, defined implicitly for each consumer i :

$$\begin{aligned} & \ln \left(\sum_{j \in J^1} \exp (x_{jt} \beta_{it} + \alpha_{it} \ln (y_i - CV_i - p_{jt}) + \xi_{jt}) \right) \\ &= \ln \left(\sum_{j \in J^0} \exp (x_{jt} \beta_{it} + \alpha_{it} \ln (y_i - p_{jt}) + \xi_{jt}) \right) \end{aligned}$$

- J^1 = with minivan; J^0 = without. Solve for CV_i numerically.

Petrin (2002): Welfare results

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984:
1982–84 CPI-ADJUSTED DOLLARS

	OLS Logit	Instrumental Variable Logit	Random Coefficients	Random Coefficients and Microdata
Compensating variation:				
Median	9,573	5,130	1,217	783
Mean	13,652	7,414	3,171	1,247

Petrin (2002): Key findings

- CV in logit model is **biased upward**
 - Due to large logit draws (extreme tastes for minivans)
- Adding micro-data reduces reliance on logit draws
- Observing actual market shares corrects Red Bus / Blue Bus bias
- **Welfare gain from minivan:** \$2.8 billion (1982 dollars)

Apple-Cinnamon Cheerios war

- Hausman (1997) in “The Economics of New Goods”
- Values welfare contribution of Apple Cinnamon Cheerios at **\$60 million per year** (mid 1990s)
- Bresnahan (1997) strongly disagrees:
 - Criticizes identifying assumptions
 - Questions assumptions about competition
- Quote: *“I have never met an economist who strayed that far from reality by ideology — only by arrogance.”*



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Monopoly pricing: setup

- **Goal:** Derive optimal pricing rule for a monopolist
- Setup:
 - Inverse demand curve: $p = P(q)$
 - Total cost: $C(q)$
 - Monopolist chooses q to maximize profit
- **Profit function:**

$$\pi(q) = \underbrace{P(q) \cdot q}_{TR} - \underbrace{C(q)}_{TC}$$

Deriving $MR = MC$

- Take first-order condition: $\frac{d\pi}{dq} = 0$

$$\frac{d}{dq} [P(q) \cdot q - C(q)] = 0$$

- Applying the product rule to $P(q) \cdot q$:

$$P(q) + P'(q) \cdot q - C'(q) = 0$$

- Rearranging:

$$\underbrace{P(q) + P'(q) \cdot q}_{MR} = \underbrace{C'(q)}_{MC}$$

- This gives us: $MR = MC$

Deriving the Lerner Index

- Start from $MR = MC$: $p + P'(q) \cdot q = MC$

- Rearrange:

$$p - MC = -P'(q) \cdot q$$

- Divide both sides by p :

$$\frac{p - MC}{p} = -P'(q) \cdot \frac{q}{p}$$

- Recall elasticity: $\varepsilon = \frac{dq}{dp} \cdot \frac{p}{q}$, so $\frac{1}{\varepsilon} = \frac{dp}{dq} \cdot \frac{q}{p} = P'(q) \cdot \frac{q}{p}$

- Therefore:

$$\boxed{\frac{p - MC}{p} = \frac{1}{|\varepsilon|}}$$

- This is the **Lerner Index**: markup equals inverse elasticity

The Lerner Index

- The **Lerner Index** measures market power:

$$L = \frac{p - MC}{p} = \frac{1}{|\varepsilon|}$$

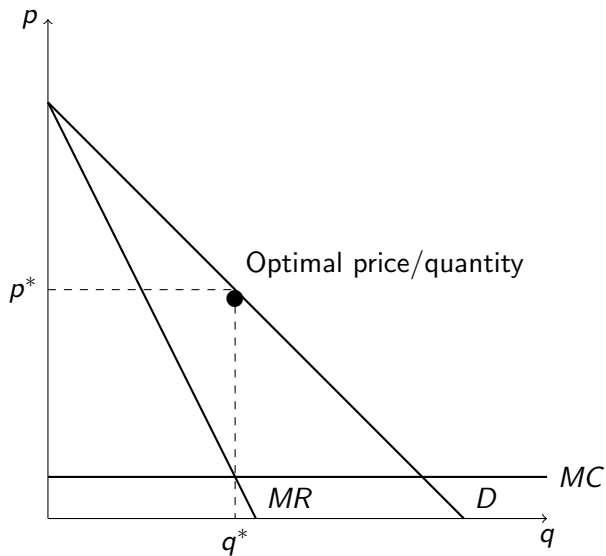
- Equivalently, solving for optimal price:

$$p = \frac{MC}{1 + \frac{1}{\varepsilon}} = \frac{MC}{1 - \frac{1}{|\varepsilon|}}$$

- **Interpretation:**

- $L = 0$: perfect competition (price = MC)
- L close to 1: high market power
- More elastic demand $\rightarrow$ lower markup

Pricing: $MR=MC$ (graph)



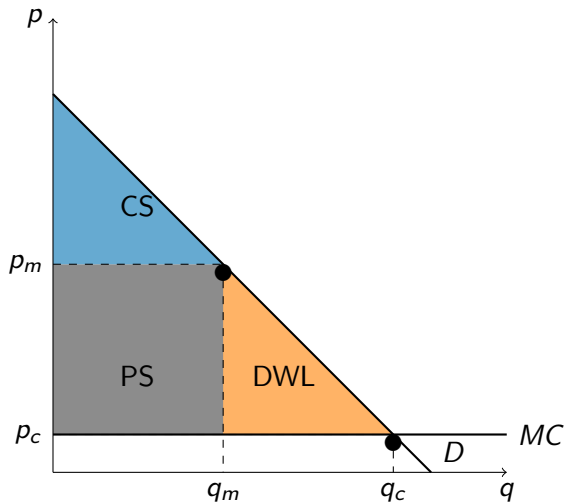
Using estimated elasticities for pricing

- Suppose you work as an economist in a firm pricing a product.
- You know MC. You **estimate** the demand elasticity ε .
- Then just plug into the elasticity rule:

$$p = \frac{MC}{1 + \frac{1}{\varepsilon}}$$

- **This is why demand estimation matters!**
 - Estimate demand $\rightarrow$ get elasticities $\rightarrow$ compute optimal prices

Welfare costs of monopoly pricing



- Competitive outcome: p_c, q_c
- Monopoly outcome: p_m, q_m
- Monopoly sets price 'too high' and quantity 'too low'
- This causes **deadweight loss** (DWL)
- Monopoly is a **market failure**

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Regulating monopolies

- How can we correct the market failure of monopoly?
- **Option 1:** Break up the monopoly (antitrust)
 - Standard Oil (1911), AT&T (1984)
 - Current debates: Google, Meta, Amazon
- Sometimes breaking up isn't possible (natural monopolies)
 - Power plants, bridges, water utilities
 - High fixed costs make one firm efficient
- How should we regulate these natural monopolies?

Regulating monopolies: case study of Facebook vs FTC



- **Background:**
- In 2012 Facebook acquired Instagram
- In 2014 Facebook acquired Whatsapp

Regulating monopolies: case study of Facebook vs FTC



- In 2020 the Federal Trade Commission (FTC) sued Facebook.
- FTC was seeking to (amongst other things) require **divestiture** (the forced sale) of Whatsapp and Instagram.
- FTC alleged “Facebook has engaged in a systematic strategy [of acquisitions]... to eliminate threats to its monopoly”
 - FTC: “Facebook’s actions to entrench and maintain its monopoly deny consumers the benefits of competition.”

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TECH

Judge dismisses FTC and state antitrust complaints against Facebook

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- **Outcome:** Court dismissed the FTC's complaint in June 2021.
- Court stated that FTC did not prove that Facebook was a monopoly.
- In addition, court argued that FTC waited too long to challenge the acquisitions.
- This and other antitrust complaints centered on large tech companies will likely continue in the near future.

Marginal cost pricing

- **Idea:** Force the monopolist to set $p = MC$
 - This is the competitive price; no deadweight loss
- **Problem:** Suppose cost is $C(q) = F + cq$ (fixed cost + marginal cost)
- If $p = MC = c$, then profit is:

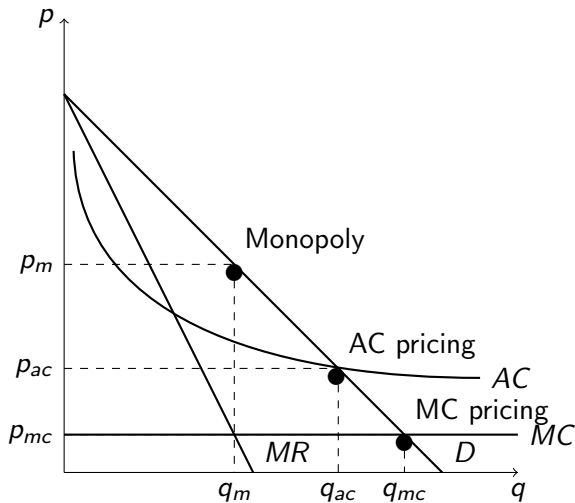
$$\pi = pq - F - cq = cq - F - cq = -F < 0$$

- Firm makes **negative** profit and would shut down!
- Solution: government **subsidy** of F

Average cost pricing

- **Alternative:** Force the monopolist to set $p = AC$
- This is the price where the monopolist makes zero profit:
 - If $\pi = 0$ then $TR = TC$
 - So $pq = TC$, which means $p = TC/q = AC$
- The monopolist can only exercise market power to cover fixed costs
- No government subsidy needed
- Commonly used for utilities (“rate of return regulation”)

Average cost pricing (graph)



Price caps

- **Price cap regulation:** Set a maximum price the firm can charge
- Often indexed to inflation (“CPI-X” regulation)
 - Price can rise with inflation, minus an efficiency factor X
- Advantage: incentivizes cost reduction
 - Under AC pricing, firm has no incentive to cut costs
 - Under price caps, cost savings become profit
- Common in UK utilities, telecoms

Key Points

1. **Log-sum formula:** $CS_{it} = \frac{1}{|\alpha_i|} \ln [\sum_j \exp(\delta_{jt} + \mu_{ijt})]$
2. **Red Bus / Blue Bus:** Logit overcounts welfare from similar new products
3. **Solution:** Use observed post-introduction shares to correct bias
4. **Lerner Index:** $\frac{p-MC}{p} = \frac{1}{|\varepsilon|}$ measures market power
5. **Monopoly regulation:** MC pricing (needs subsidy), AC pricing (zero profit), price caps
6. More elastic demand $\rightarrow$ lower markup